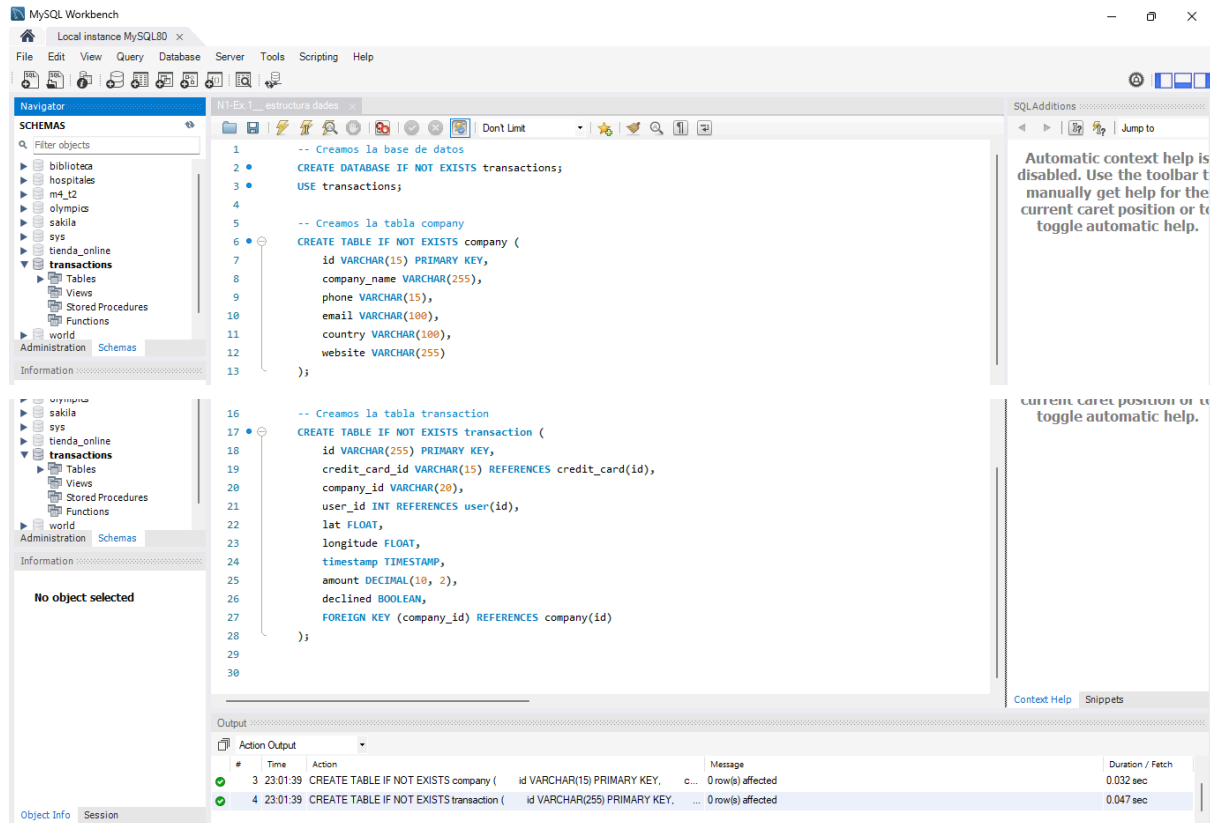


NIVELL 1

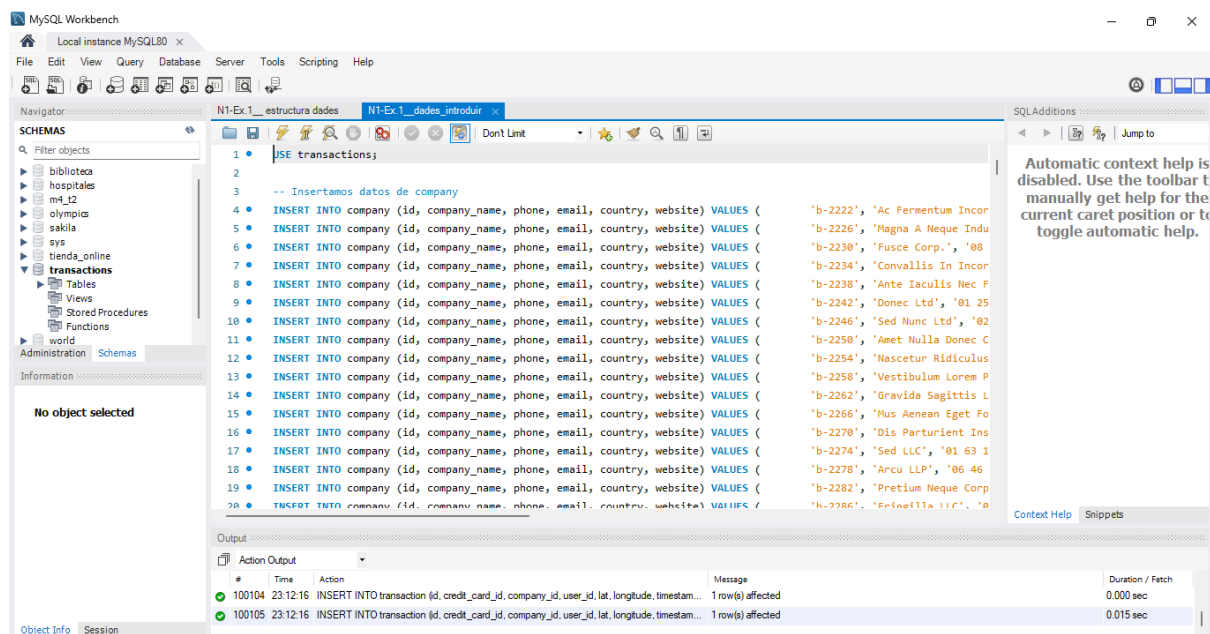
Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

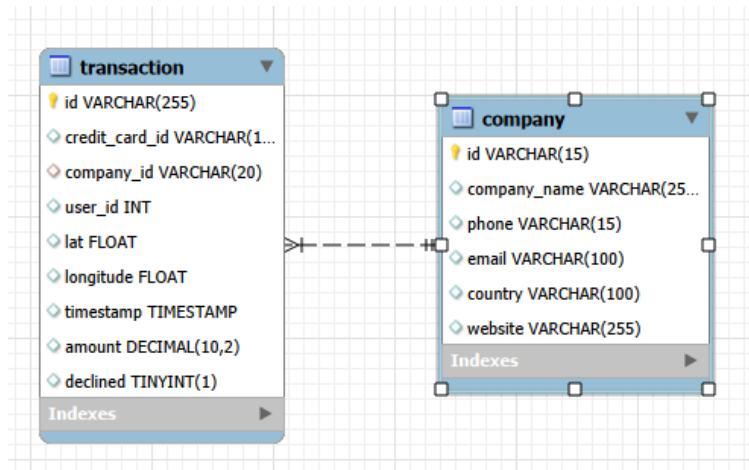
- Importo estructura_dades (File > Open SQL Script)



- Importo dades_introduir (File > Open SQL Script)



- Características principales del esquema y Explicación de tablas y variables



La presente base de datos está formada por dos tablas, “company” y “transactions”. Nos encontramos ante un esquema de tipo estrella. Este esquema se caracteriza por tener una tabla de hechos (transactions) y una o más de dimensiones (company), pero sin “subdimensiones” (que lo convertiría en un esquema de copo de nieve). En cuanto a relaciones, podemos concluir que nos encontramos ante un modelo 1 a n, donde una compañía puede estar presente en varias transacciones.

En la tabla “**company**”:

Field	Type	Null	Key	Default	Extra
id	varchar(15)	NO	PRI		
company_name	varchar(255)	YES			
phone	varchar(15)	YES			
email	varchar(100)	YES			
country	varchar(100)	YES			
website	varchar(255)	YES			

Observamos 6 columnas distintas, todas ellas contienen variables de tipo VARCHAR (distinta capacidad, de 15 a 255). Son: “id”, “company_name”, “phone”, “email”, “country” y “website”.

Todas ellas aceptan valores NULL a excepción de “id”, que es la PRIMARY KEY.

Para comprobar cuantos registros incluye, podemos observar las filas que retorna el output al ejecutar `SELECT *` o utilizar `SELECT COUNT(*)`, y ambas resultan 100 filas.

MySQL Workbench interface showing a query executed on the 'company' table. The query is: `SELECT * FROM transactions.company;`

The result grid displays the following data:

id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.co.uk	Germany	https://cnn.com/user/110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netfix.com/settings
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.co.uk	Norway	https://nytimes.com/user/110
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one
b-2250	Amet Nulla Donec Corporation	07 15 25 14 74	mattis.integer.eu@protonmail.net	Italy	https://netfix.com/sub/cars
b-2254	Nascetur Ridiculus Mus Inc.	06 26 87 61 84	suspendisse.dui@icloud.net	United States	https://ebay.com/sub
b-2258	Vestibulum Lorem PC	02 02 87 33 40	aenean.mass@integer@aol.net	Belgium	https://pinterest.com/sub/cars
b-2262	Gravida Sagittis LLP	03 81 28 33 97	turpis.vitae@google.ca	Sweden	https://naver.com/site
b-2266	Mus Aenean Eget Foundation	06 25 15 52 43	mi.duis@hotmail.net	Sweden	https://instagram.com/group/9
b-2270	Dis Parturient Inconthite	05 36 79 78 74	risus@integerprotonmail.com	Ireland	https://linnovate.com/one

The output shows the query completed successfully, returning 100 rows.

MySQL Workbench interface showing a query executed on the 'company' table. The query is: `SELECT COUNT(*) FROM transactions.company;`

The result grid displays the following data:

id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.co.uk	Germany	https://cnn.com/user/110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netfix.com/settings
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.co.uk	Norway	https://nytimes.com/user/110
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one
b-2250	Amet Nulla Donec Corporation	07 15 25 14 74	mattis.integer.eu@protonmail.net	Italy	https://netfix.com/sub/cars
b-2254	Nascetur Ridiculus Mus Inc.	06 26 87 61 84	suspendisse.dui@icloud.net	United States	https://ebay.com/sub
b-2258	Vestibulum Lorem PC	02 02 87 33 40	aenean.mass@integer@aol.net	Belgium	https://pinterest.com/sub/cars
b-2262	Gravida Sagittis LLP	03 81 28 33 97	turpis.vitae@google.ca	Sweden	https://naver.com/site
b-2266	Mus Aenean Eget Foundation	06 25 15 52 43	mi.duis@hotmail.net	Sweden	https://instagram.com/group/9
b-2270	Dis Parturient Inconthite	05 36 79 78 74	risus@integerprotonmail.com	Ireland	https://linnovate.com/one

The output shows the query completed successfully, returning 1 row.

En la tabla "transaction":

MySQL Workbench interface showing a query executed on the 'transaction' table. The query is: `DESCRIBE transaction;`

The result grid displays the following data:

Field	Type	Null	Key	Default	Extra
id	varchar(255)	NO	PRI		
credit_card_id	varchar(15)	YES			
company_id	varchar(20)	YES	MUL		
user_id	int	YES			
lat	float	YES			
longitude	float	YES			
timestamp	timestamp	YES			
amount	decimal(10,2)	YES			
declined	tinyint(1)	YES			

The output shows the query completed successfully, returning 9 rows.

Observamos 9 columnas distintas: “id”, “credit_card_id” y “company_id”, que contienen variables de tipo varchar, “user_id” (tipo int -números enteros-), “lat” y “longitude” (ambas con variables tipo float -números decimales-), “timestamp” (tipo timestamp -combinación de fecha y hora-), “amount” (tipo decimal -números decimales más preciso que float-) y “declined” (tipo tinyint -número entero más bien pequeño-).

Todas ellas aceptan valores NULL a excepción de “id”, que es la PRIMARY KEY.

Además, la columna “company-id” actúa como FOREIGN KEY

Para contar registros, igual que antes, podemos emplear varios métodos. En este caso, el resultado es 100000:

The screenshot shows the MySQL Workbench interface. The SQL editor contains the query: `SELECT * FROM transactions.transaction;`. The result grid displays 10 rows of data with columns: id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, and declined. The bottom status bar indicates 'Query Completed'.

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00043A49-2949-494B-A5D0-A5BAE38B19D0	CcS-9294	b-2458	4713	46.1999	1.43554	2024-08-28 07:16:46	395.43	0
00047FE-8650-40CF-830E-C7ED0EE1CAAD	CcS-5019	b-2370	438	41.5972	12.2218	2016-12-21 20:07:18	155.63	0
00043068-ED2E-4F2F-8186-CEE074D875D0	CcS-6699	b-2390	2118	29.7573	-95.3796	2020-07-14 15:37:45	326.01	0
000481C3-1C26-4FEF-8340-4C0EB0049BD	CcS-6696	b-2230	2115	53.5489	-113.503	2017-09-04 19:44:53	161.60	0
00051AA4-9C8E-4268-8070-C38062A183E2	CcS-7605	b-2265	3025	52.2084	5.69081	2017-01-05 18:19:25	148.01	0
0008A312-EDFE-4A4F-8C99-E9C92EC3CA4D	CcJ-3358	b-2598	215	53.5535	-113.499	2023-09-23 04:51:43	294.59	0
0009A151-98CF-4E31-9053-A468FF7FAAB	CcS-7509	b-2546	2928	51.9362	5.34265	2023-12-31 00:06:36	383.63	0
00090494-6245-4DF9-95D0-2C084191CFFB	CcS-8483	b-2526	3902	45.492	-73.5706	2017-07-18 07:52:02	197.80	0
000A1DEC-CD86-4AB2-A619-71DAB8D4A262	CcS-6467	b-2558	1886	55.7425	-3.30009	2018-09-08 05:29:58	339.94	0

The screenshot shows the MySQL Workbench interface. The SQL editor contains the query: `SELECT COUNT(*) FROM transactions.transaction;`. The result grid displays a single row with the value 100000. The bottom status bar indicates 'Query Completed'.

COUNT(*)
100000

Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 # Exercici 2
2
3 SELECT DISTINCT c.country
4 FROM company AS c
5 JOIN transaction AS t
6 ON c.id = t.company_id
7 ORDER BY c.country;
```

The Results tab shows the following data:

country
Australia
Belgium
Canada
China
France
Germany
Ireland
Italy
Netherlands

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
100113	23:28:25	SELECT COUNT(*) FROM transactions.transaction	1 row(s) returned	0.062 sec / 0.000 sec
100114	23:40:44	SELECT DISTINCT c.country FROM company AS c JOIN transaction AS t ON c.id = t.co...	15 row(s) returned	0.094 sec / 0.000 sec

- Empleo DISTINCT para evitar duplicar países
- Por un tema estético / facilidad de lectura, ordeno los resultados alfabéticamente (ORDER BY lo hace así por defecto)
- No pongo alias (con AS) a la columna por tener un nombre suficientemente “limpio” e intuitivo

- Des de quants països es generen les vendes.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 # Exercici 2
2
3 SELECT COUNT(DISTINCT c.country) AS num_paisos
4 FROM company AS c
5 JOIN transaction AS t
6 ON c.id = t.company_id;
```

The Results tab shows the following data:

num_paisos
15

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
100114	23:40:44	SELECT DISTINCT c.country FROM company AS c JOIN transaction AS t ON c.id = t.co...	15 row(s) returned	0.094 sec / 0.000 sec
100115	23:47:25	SELECT COUNT(DISTINCT c.country) AS num_paisos FROM company AS c JOIN trans...	1 row(s) returned	0.313 sec / 0.000 sec

- Empleo DISTINCT para evitar duplicar países
- Uso INNER JOIN para “eliminar” las empresas que no han hecho ventas
- Establezco alias con AS para mejor comprensión

- **Identifica la companyia amb la mitjana més gran de vendes.**

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

2 SELECT c.company_name, ROUND(AVG(t.amount),2) AS mitjana_vendes
3 FROM company AS c
4 JOIN transaction AS t
5 ON c.id = t.company_id
6 GROUP BY c.company_name
7 ORDER BY mitjana_vendes DESC
8 LIMIT 1;

```

The Result Grid shows the following data:

company_name	mitjana_vendes
Ac Fermentum Incorporated	284.87

The Output tab shows the execution details:

#	Time	Action	Message	Duration / Fetch
100115	23:47:25	SELECT COUNT(DISTINCT c.country) AS num_paises FROM company AS c JOIN trans...	1 row(s) returned	0.313 sec / 0.000 sec
100116	23:52:27	SELECT c.company_name, ROUND(AVG(t.amount),2) AS mitjana_vendes FROM compa...	1 row(s) returned	0.687 sec / 0.000 sec

- Empleo ROUND para redondear decimales y así facilitar la legibilidad, además, porque aunque matemáticamente pueda haber infinitos decimales, en nuestro sistema económico nos detenemos en el céntimo (2)
- ORDER BY (especificando DESC) + LIMIT 1 permiten señalar la más alta

Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

- **Mostra totes les transaccions realitzades per empreses d'Alemanya.**

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

3 SELECT t.id
4 FROM transaction AS t
5 WHERE t.company_id IN (
6 SELECT c.id
7 FROM company AS c
8 WHERE c.country = "Germany"
9 );

```

The Result Grid shows the following data:

id
00138038-206D-4C03-94B7-63A2676EB9B4
0013C1B6-3894-4D5C-8154-E2B3FEB0A8E9
00201A11-2652-44C4-941D-198FC80E77F0
00235618-0A5C-4D49-9DCB-83A9405D8923
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38
00687139-48B2-4FFA-8E73-820376F04AB4
0074F4D0-32F1-4827-8758-55896314623A
00AAB9CD-3906-4DCB-8A1D-138E73DC90A9
00BE09D4-6920-47D8-ABE8-325E2269829D

The Output tab shows the execution details:

#	Time	Action	Message	Duration / Fetch
100116	23:52:27	SELECT c.company_name, ROUND(AVG(t.amount),2) AS mitjana_vendes FROM compa...	1 row(s) returned	0.687 sec / 0.000 sec
100117	23:56:13	SELECT t.id FROM transaction AS t WHERE t.company_id IN (SELECT c.id FROM ...	13291 row(s) returned	0.000 sec / 0.015 sec

- La subconsulta encuentra las empresas de Alemania
- La consulta principal, las transacciones realizadas por el resultado de la subconsulta

- **Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.**

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

3  SELECT co.id, co.company_name
4  FROM company AS co
5  WHERE EXISTS (
6    SELECT 1
7    FROM transaction AS tr
8    WHERE tr.company_id = co.id
9    AND tr.amount > (
10     SELECT AVG(amount)
11     FROM transaction AS tr1)
12 );

```

The Result Grid shows the following data:

id	company_name
b-2222	Ac Fermentum Incorporated
b-2226	Magna A Neque Industries
b-2230	Fusce Corp.
b-2234	Convallis In Incorporated
b-2238	Ante Iaculis Nec Foundation
b-2242	Donec Ltd

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
100117	23:56:13	SELECT t.id FROM transaction AS t WHERE t.company_id IN (SELECT c.id FROM ...	13291 row(s) returned	0.000 sec / 0.015 sec
100118	00:02:26	SELECT co.id, co.company_name FROM company AS co WHERE EXISTS (SELECT 1 ...	100 row(s) returned	0.250 sec / 0.000 sec

- La consulta principal tiene como condición que la empresa esté en el resultado de la subconsulta
- La primera subconsulta escogerá según la mediana sea superior al resultado de la segunda subconsulta
- La segunda subconsulta calcula la media de amount
- **Eliminarán del sistema las empresas que no tienen transacciones registradas, entrega el llistat d'aquestes empreses.**

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

1  # Exercici 3
2
3  SELECT c.company_name
4  FROM company AS c
5  WHERE NOT EXISTS (
6    SELECT 1
7    FROM transaction AS t2
8    WHERE t2.company_id = c.id
9  )
10 # Totes les empreses han registrat transaccions

```

The Result Grid shows the following data:

company_name
company 12

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
100122	00:12:16	SELECT c.company_name FROM company AS c WHERE NOT EXISTS (SELECT DI...	0 row(s) returned	0.000 sec / 0.000 sec
100123	00:15:20	SELECT c.company_name FROM company AS c WHERE NOT EXISTS (SELECT 1 ...	0 row(s) returned	0.016 sec / 0.000 sec

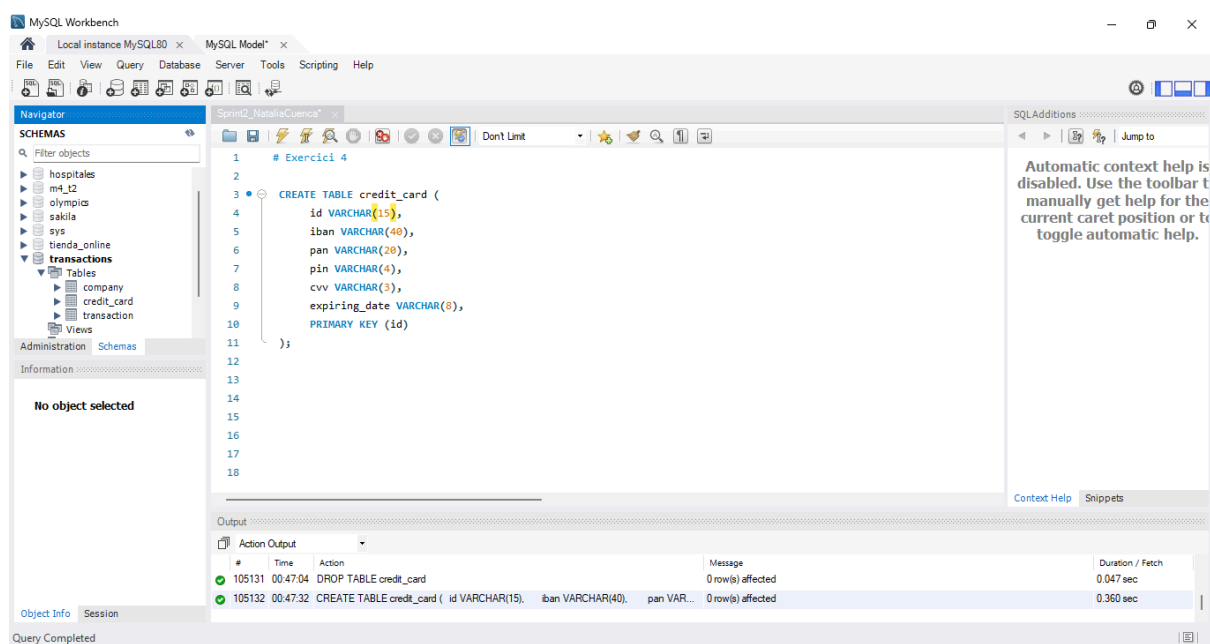
- Empleo NOT EXISTS en vez de NOT IN ya que el primero “no se fija” en los NULL y el segundo sí (aunque en este caso dan el mismo resultado).

- Al no devolver registros, concluimos que no hay empresas que no hayan registrado transacciones

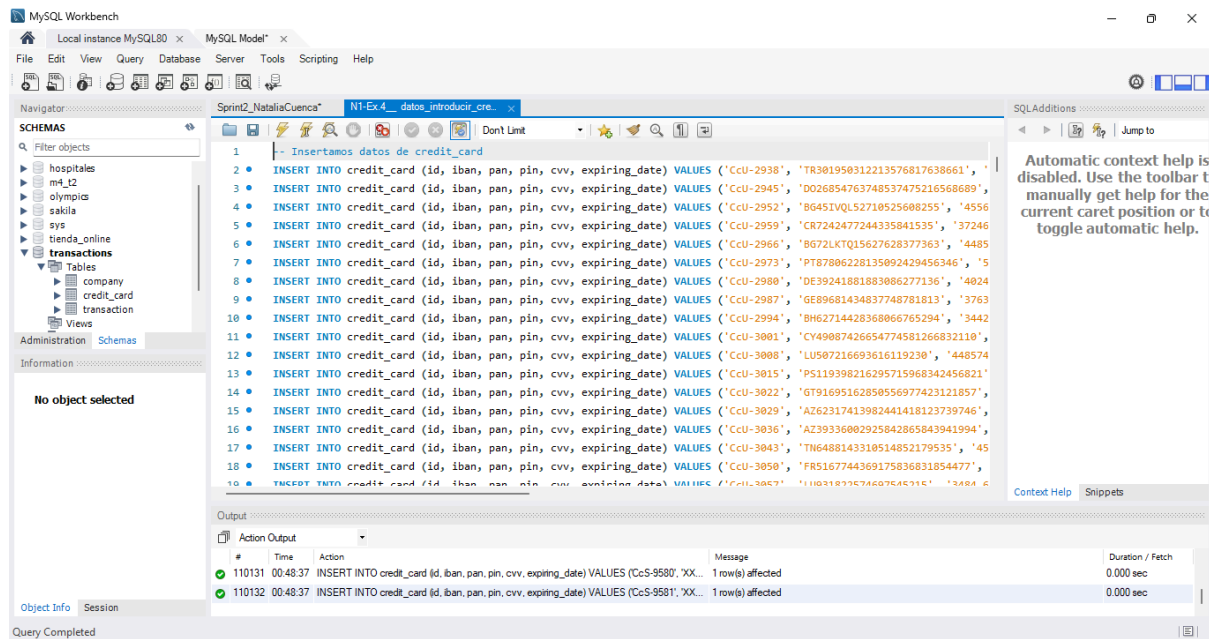
Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

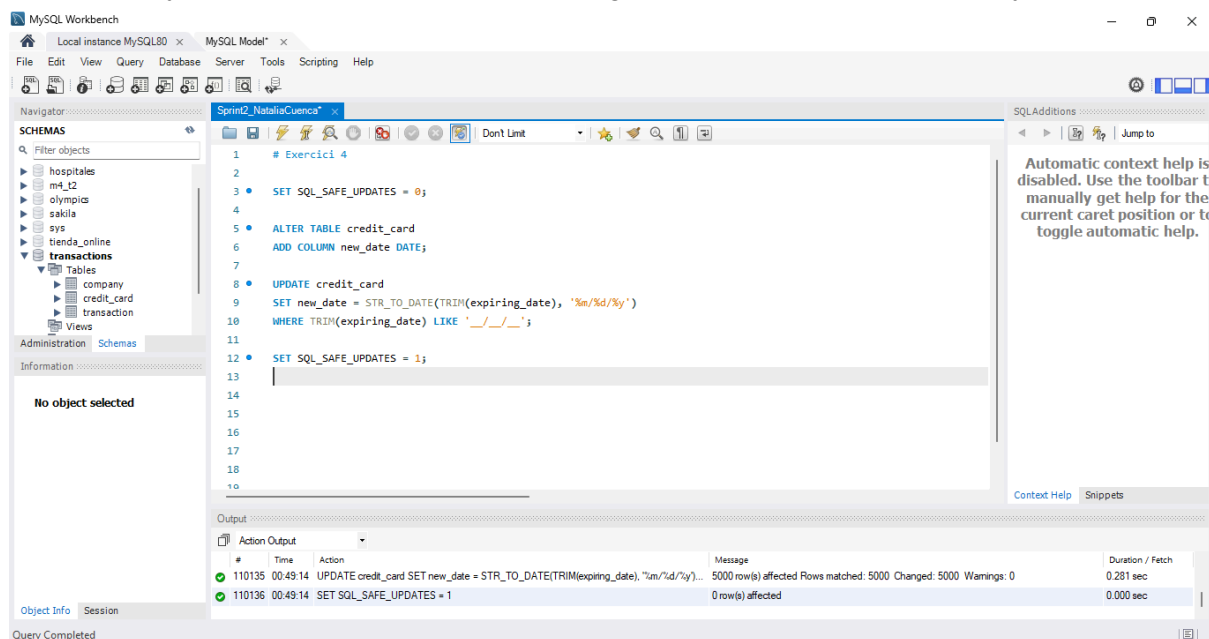
- Creo la tabla:

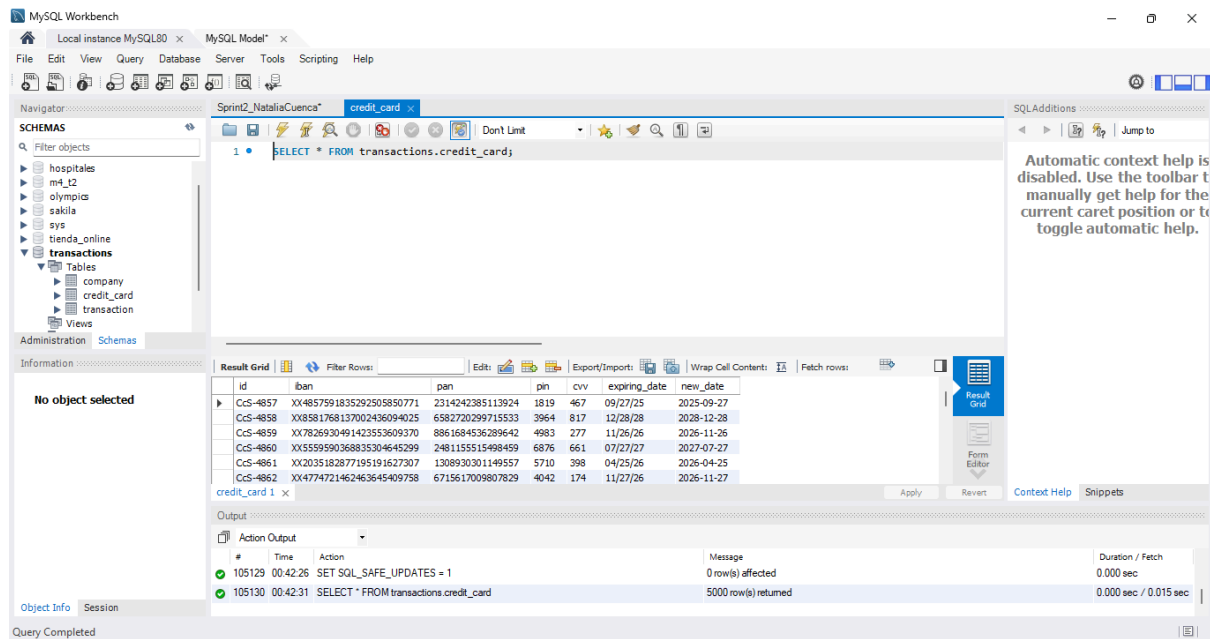


- Inserto los datos del archivo previamente descargado (File > Open SQL Script)



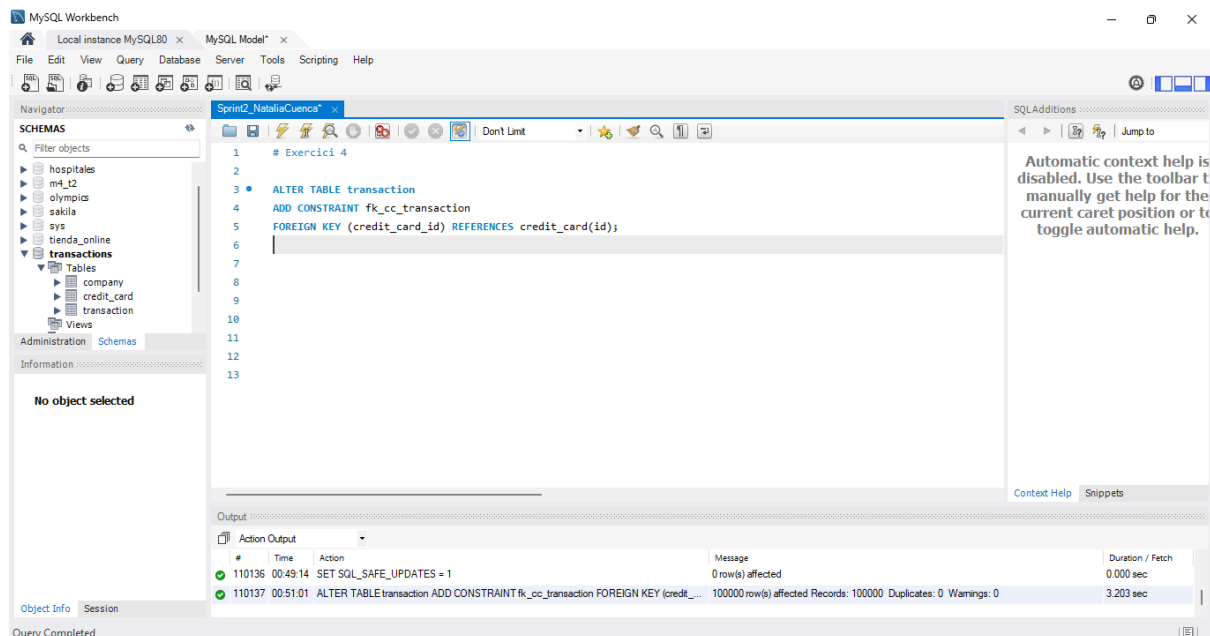
- Una vez me aseguro de que los datos se han importado correctamente, como la fecha está en un formato complejo, la estandarizo creando una nueva columna. No borro la vieja por trazabilidad:
- SET SQL_SAFE_UPDATES es la manera de desactivar y activar (en este caso, temporalmente) filtros de seguridad que vienen por defecto en las últimas versiones de MySQL Workbench que dificultan algunas modificaciones de bbdd y tablas



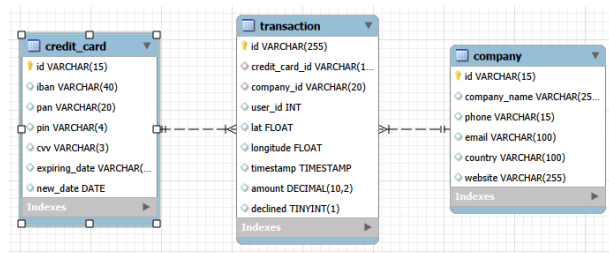


(En esta imagen se pueden ver ambas columnas)

- A continuación creo la relación entre tablas



- Características principales del esquema



A las dos tablas anteriores, “company” y “transactions”, hemos añadido una tercera “credit_card”. Seguimos ante un esquema de tipo estrella.

En cuanto a relaciones, podemos concluir que nos encontramos ante un modelo 1 a n en ambos casos, donde una compañía puede estar presente en varias transacciones y una tarjeta de crédito puede realizar varias transacciones.

Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és:

TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

- Primero hago SELECT para ver la situación actual

MySQL Workbench interface showing a query and its results.

Query:

```
1 # Exercici 5
2
3 SELECT *
4 FROM credit_card
5 WHERE id = 'CcU-2938';
6
7
8
9
10
11
```

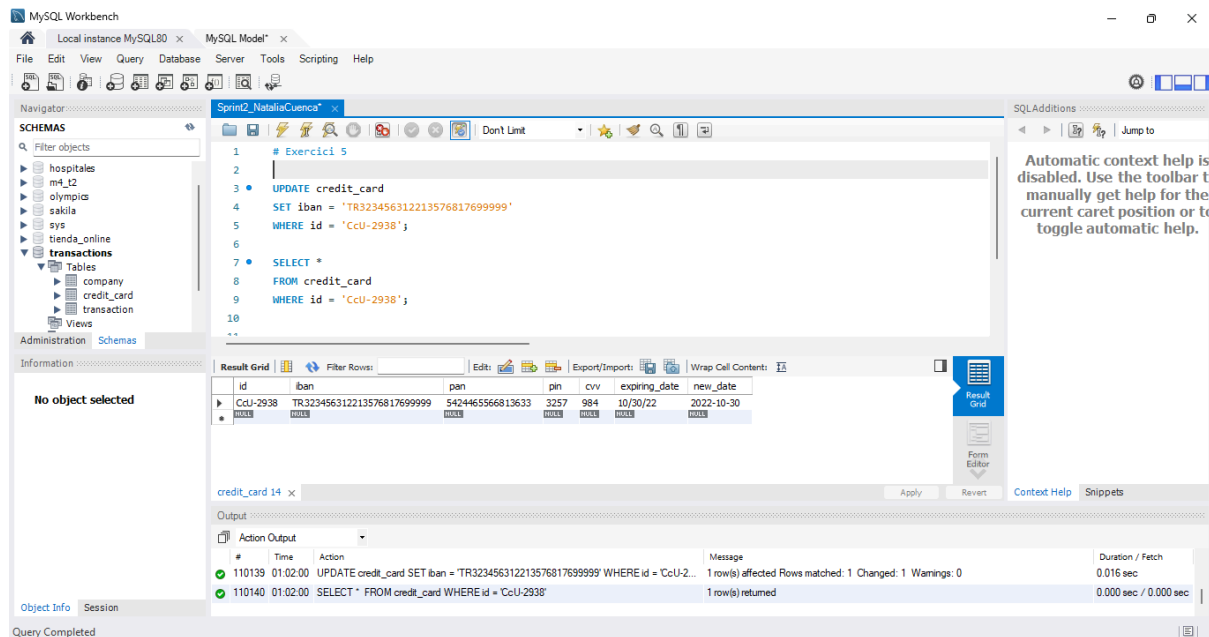
Result Grid:

id	iban	pan	pin	cvv	expiring_date	new_date
CcU-2938	TR301950312213576817638661	5424465566813633	3257	984	10/30/22	2022-10-30

Output:

#	Time	Action	Message	Duration / Fetch
110137	00:51:01	ALTER TABLE transaction ADD CONSTRAINT fk_cc_transaction FOREIGN KEY (credit...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	3.203 sec
110138	01:00:46	SELECT * FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.015 sec / 0.000 sec

- Modifico la información con UPDATE y compruebo con SELECT de nuevo



Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

Id 108B1D1D-5B23-A76C-55EF-C568E49A99DD

credit_card_id CcU-9999

company_id b-9999

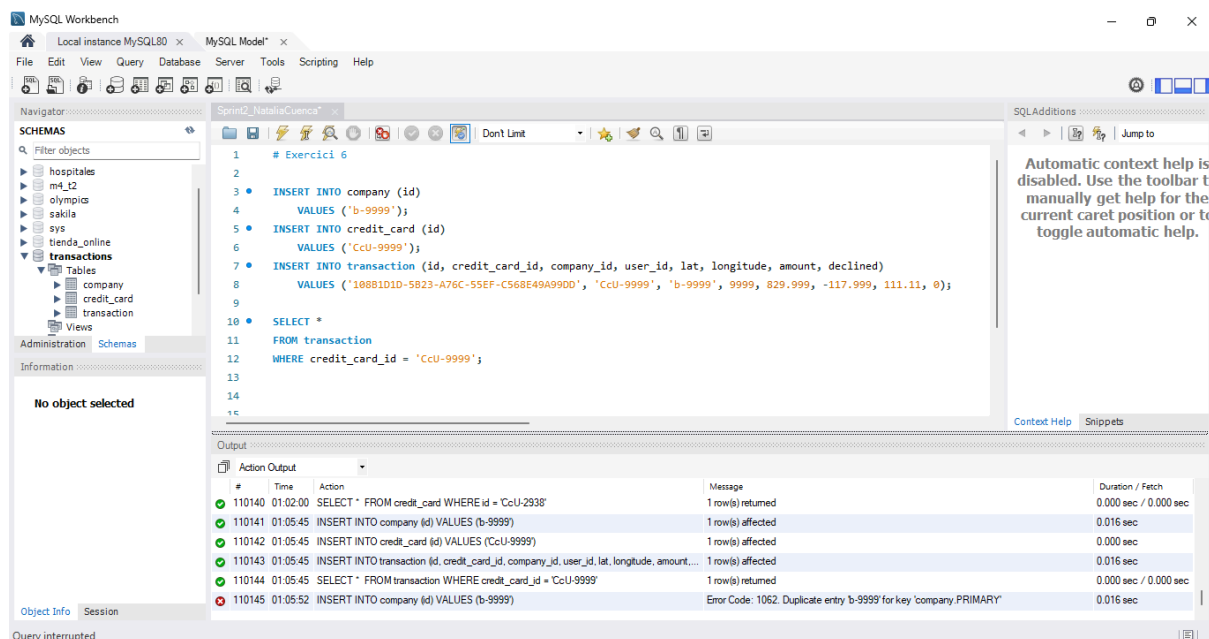
user_id 9999

lat 829.999

longitude -117.999

amount 111.11

declined 0



- Creo los ID en ambas tablas y, seguidamente, el resto de información. Compruebo:

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

7 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
8   VALUES ('10881010-5823-A76C-55EF-C568E49A9DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);
9
10 • SELECT *
11   FROM transaction
12  WHERE credit_card_id = 'CcU-9999';
13

```

The Results tab shows the output of the query:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
1088 ID ID-5823-A76C-55EF-C568E49A9DD	CcU-9999	b-9999	9999	829.999	-117.999	111.11	0	

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
110141	01:05:45	INSERT INTO company (id) VALUES (b-9999)	1 row(s) affected	0.016 sec
110142	01:05:45	INSERT INTO credit_card (id) VALUES (CcU-9999)	1 row(s) affected	0.000 sec
110143	01:05:45	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)	1 row(s) affected	0.016 sec
110144	01:05:45	SELECT * FROM transaction WHERE credit_card_id = 'CcU-9999'	1 row(s) returned	0.000 sec / 0.000 sec
110145	01:05:52	INSERT INTO company (id) VALUES (b-9999)	Error Code: 1062. Duplicate entry 'b-9999' for key 'company.PRIMARY'	0.016 sec
110146	01:07:28	SELECT * FROM transaction WHERE credit_card_id = 'CcU-9999'	1 row(s) returned	0.000 sec / 0.000 sec

Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordra mostrar el canvi realitzat.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

1 # Exercici 7
2
3 • ALTER TABLE credit_card
4   DROP COLUMN pan;
5
6 • SELECT *
7   FROM credit_card;

```

The Results tab shows the output of the query:

id	iban	pin	cvv	expiring_date	new_date
CcS-4857	XX4857591835292505850771	1819	467	09/27/25	2025-09-27
CcS-4858	XX8581768137002436094025	3964	817	12/28/28	2028-12-28
CcS-4859	XX7826930491423553609370	4983	277	11/26/26	2026-11-26
CcS-4860	XX5559590368835304645299	6876	661	07/27/27	2027-07-27
CcS-4861	XX2035182877195191627307	5710	398	04/25/26	2026-04-25

The Output tab shows the execution log:

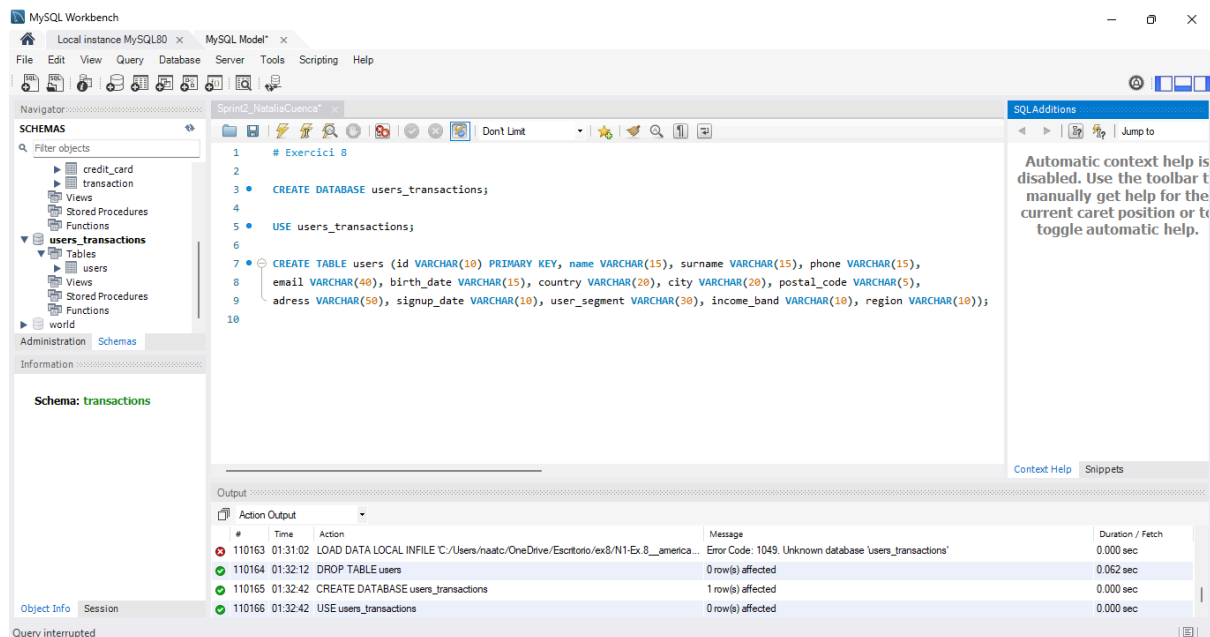
#	Time	Action	Message	Duration / Fetch
110143	01:05:45	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)	1 row(s) affected	0.016 sec
110144	01:05:45	SELECT * FROM transaction WHERE credit_card_id = 'CcU-9999'	1 row(s) returned	0.000 sec / 0.000 sec
110145	01:05:52	INSERT INTO company (id) VALUES (b-9999)	Error Code: 1062. Duplicate entry 'b-9999' for key 'company.PRIMARY'	0.016 sec
110146	01:07:28	SELECT * FROM transaction WHERE credit_card_id = 'CcU-9999'	1 row(s) returned	0.000 sec / 0.000 sec
110147	01:09:08	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.125 sec
110148	01:09:08	SELECT * FROM credit_card	5001 row(s) returned	0.016 sec / 0.000 sec

Exercici 8

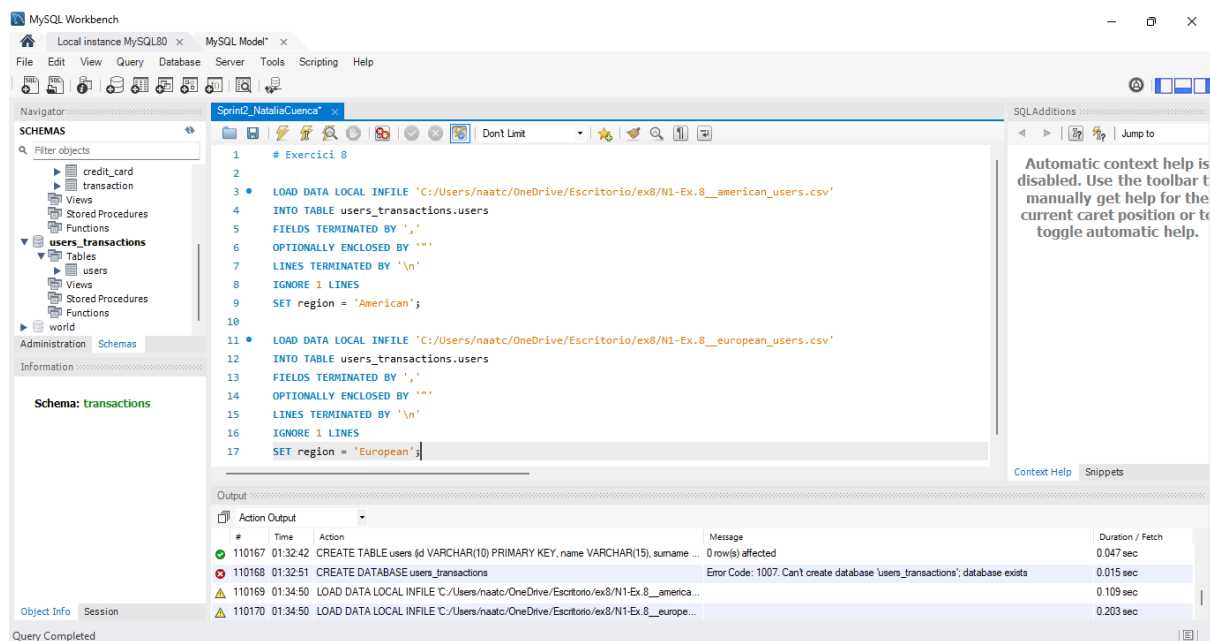
Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american_users.csv
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

**Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguem realitzar les següents consultes:
La taula de products.csv l'utilitzarem més endavant.**



- Creo la bbdd que nombro users_transactions
- Tras revisar los csv, veo que american_users y european_users se pueden juntar en una sola tabla (no hay id repetidos y contienen la misma información). Creo la tabla estableciendo el id como PK y, para no perder la información, añado una columna que se llama "region" para saber de que tabla proviene cada usuario
- Me dispongo a seguir creando las tablas e importando los datos, comprobando que se ha añadido correctamente



id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	Zeus	Gamble	1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sag
10	Robert	McCarthy	(324) 746-6771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773
100	Melodie	Mdean	1-677-221-7152	risus.varius@google.ca	Sep 15, 1989	United States	San Jose	95101	Ap #644-849
1000	Amikrv	Qbuixbp	+48-258-9936	amikrv.qbuixbp@example.com	May 17, 1970	Germany	Stuttgart	70173	215 Qbuixbp
1001	Nfrib	Oydaibvg	+94-121-2522	nfrib.oydaibvg@example.com	Mar 4, 1994	Germany	Cologne	50667	121 Oydaibvg

```
1 # Ejercicio 8
2
3 CREATE TABLE companies (company_id VARCHAR(10) PRIMARY KEY, company_name VARCHAR(50), phone VARCHAR(15), email VARCHAR(50),
4 country VARCHAR(20), website VARCHAR(40), merchant_category VARCHAR(15), merchant_price VARCHAR(15),
5 merchant_price_position VARCHAR(15));
6
7 LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_companies.csv'
8 INTO TABLE users_transactions.companies
9 FIELDS TERMINATED BY ','
10 OPTIONALLY ENCLOSED BY '"'
11 LINES TERMINATED BY '\n'
12 IGNORE 1 LINES;
13
14
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

#	Time	Action	Message	Duration / Fetch
110170	01:34:50	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_europe...		0.203 sec
110171	01:34:56	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_america...		0.015 sec
110172	01:34:56	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_europe...		0.047 sec
110173	01:35:41	SELECT * FROM users_transactions.users	5000 row(s) returned	0.000 sec / 0.016 sec

Query Completed

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	+64 601 244 632	donec.porttitor.tellus@yahoo.net	Germany	https://www.ac-fermentum-incorporate...
b-2226	Magna A Neque Industries	+33 823 560 377	risus.donec.nibh@icloud.org	Australia	https://www.magna-a-neque-industries...
b-2230	Fusce Corp.	+29 657 646 432	risus@protonmail.edu	United States	https://www.fusce-corp.com
b-2234	Convalis In Incorporated	+6 693 982 1255	mauris.ut@aol.co.uk	Germany	https://www.convalis-in-incorporated.c...
b-2238	Ante Taculis Nec Foundation	+69 914 549 100	sed.dictum.pron@outlook.ca	New Zealand	https://www.ante-taculis-nec-foundatio...

```
1 # Ejercicio 8
2
3 CREATE TABLE credit_cards (id VARCHAR(10) PRIMARY KEY, user_id VARCHAR(50), iban VARCHAR(30), pan VARCHAR(20),
4 pin VARCHAR(5), cvv VARCHAR(3), track1 VARCHAR(50), track2 VARCHAR(50), expiring_date VARCHAR(15),
5 card_type VARCHAR(15), card_renewal_flag VARCHAR(1));
6
7 LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_credit_cards.csv'
8 INTO TABLE users_transactions.credit_cards
9 FIELDS TERMINATED BY ','
10 OPTIONALLY ENCLOSED BY '"'
11 LINES TERMINATED BY '\n'
12 IGNORE 1 LINES;
13
14
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

#	Time	Action	Message	Duration / Fetch
110174	01:36:55	CREATE TABLE companies (company_id VARCHAR(10) PRIMARY KEY, company_name...	0 row(s) affected	0.031 sec
110175	01:36:55	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex_8_compa...	100 row(s) affected, 214 warning(s): 1265 Data truncated for column 'phone' at row 1 126...	0.047 sec
110176	01:37:34	SELECT * FROM users_transactions.users	5000 row(s) returned	0.000 sec / 0.015 sec
110177	01:37:58	SELECT * FROM users_transactions.companies	100 row(s) returned	0.000 sec / 0.000 sec

Query Completed

id	user_id	iban	pan	pin	cvv	track1	track2
CS-4657	276	ES415679416676968711699208	2314242385113924	1819	467	%B2314242385113924*UWCBUDUWCBUD*22...	%B23142424...
CS-4658	277	NL252033037609851262242244	6582720299715533	3964	817	%B6582720299715533*TTQWMTTQWMT*2404...	%B65827202...
CS-4659	278	SE720084318584364771497262	8861684536289642	4983	277	%B8861684536289642*CORBODCORBOD*280...	%B8861684...
CS-4660	279	ES165574078045314365283427	2481155515498459	6876	661	%B2481155515498459*TLTUTLUTLUT*31040...	%B2481155...
CS-4661	280	SE656762210431025196620841	1308930301149557	5710	398	%B1308930301149557*HPOBNZ*HPOBNZ*330...	%B1308930...

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a filter for 'users_transactions'. The main editor shows a SQL script for 'Exercici 8' that creates a table named 'transaction1' and loads data from a CSV file. The output pane shows the execution results of the queries.

SQL Script:

```

1 # Exercici 8
2
3 CREATE TABLE transaction1 (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(10), bussines_id VARCHAR(10), timestamp VARCHAR
4 amount VARCHAR(10), declined VARCHAR(1), product_id VARCHAR(5), user_id VARCHAR(10), lat VARCHAR(20),
5 longitude VARCHAR(20), discount_amount VARCHAR(10), tax_amount VARCHAR(10), shipping_amount VARCHAR(10),
6 channel VARCHAR(20), campaign_id VARCHAR(20), device_type VARCHAR(10), is_international VARCHAR(1),
7 decline_reason VARCHAR(20), distance_km VARCHAR(10));
8
9 LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex-8_transactions.csv'
10 INTO TABLE users_transactions.transaction1
11 FIELDS TERMINATED BY ','
12 OPTIONALLY ENCLOSED BY '"'
13 LINES TERMINATED BY '\n'
14 IGNORE 1 LINES;
15

```

Output:

#	Time	Action	Message	Duration / Fetch
110177	01:37:58	SELECT * FROM users_transactions.companies	100 row(s) returned	0.000 sec / 0.000 sec
110178	01:38:55	CREATE TABLE credit_cards (id VARCHAR(10) PRIMARY KEY, user_id VARCHAR(50), i...	0 row(s) affected	0.031 sec
110179	01:38:55	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex-8_cred...		0.203 sec
110180	01:39:12	SELECT * FROM users_transactions.credit_cards	5000 row(s) returned	0.000 sec / 0.016 sec

Result Grid:

id	card_id	bussines_id	timestamp	amount	declined	product_id	user_id	lat
00043A49-2949-494B-A5D0-ASBAE38B19D0	CcS-9294	b-2458	2024-08-28 07:16:46	433.51	0	16, 2	4713	46.1999292615
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	CcS-5019	b-2370	2016-12-21 20:07:18	172.56	0	66, 6	438	41.5972055446
00045D68-ED2E-4F2F-8186-CEE074D875D0	CcS-6699	b-2390	2020-07-14 15:37:45	332.38	0	30, 1	2118	29.7572958996
000481C3-1C26-4FEF-83A0-4CD0EB004BBD	CcS-6696	b-2230	2017-09-04 19:44:53	175.00	0	72	2115	53.5488837675
00051AA4-9CBE-4268-B070-C38062A183E2	CcS-7606	b-2266	2017-01-05 18:19:25	158.55	0	18	3025	52.2083695165

- Seguidamente creo las relaciones entre tablas (FK)

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a filter for 'users_transactions'. The main editor shows a SQL script for 'Exercici 8' that creates foreign key constraints for the 'transaction1' table. The output pane shows the execution results of the queries.

SQL Script:

```

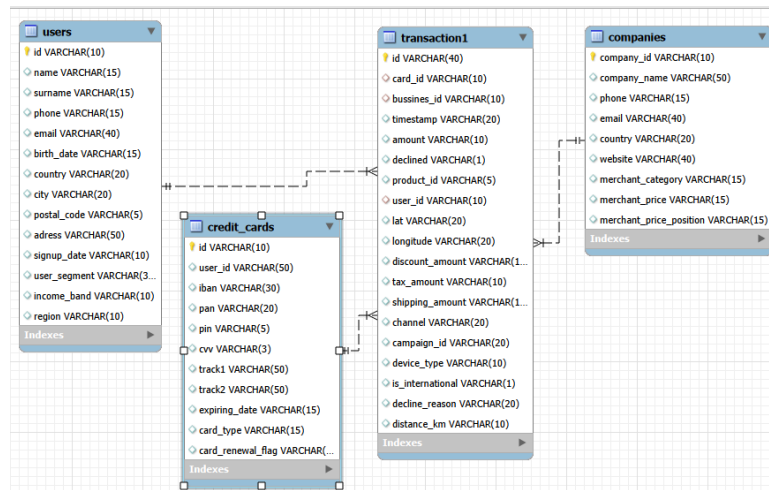
1 # Exercici 8
2
3 ALTER TABLE transaction1
4 ADD CONSTRAINT fk_transactions_companies
5 FOREIGN KEY (bussines_id) REFERENCES companies(company_id),
6 ADD CONSTRAINT fk_transactions_users
7 FOREIGN KEY (user_id) REFERENCES users(id),
8 ADD CONSTRAINT fk_transactions_cards
9 FOREIGN KEY (card_id) REFERENCES credit_cards(id);

```

Output:

#	Time	Action	Message	Duration / Fetch
110179	01:38:55	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex-8_cred...		0.203 sec
110180	01:39:12	SELECT * FROM users_transactions.credit_cards	5000 row(s) returned	0.000 sec / 0.016 sec
110181	01:41:41	CREATE TABLE transaction1 (id VARCHAR(40) PRIMARY KEY, card_id VARCHAR(10), ...	0 row(s) affected	0.031 sec
110182	01:41:41	LOAD DATA LOCAL INFILE 'C:/Users/naatc/OneDrive/Escritorio/ex8/N1-Ex-8_transac...		3.922 sec

- Y este es el diagrama resultante:



Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```

1  USE users_transactions;
2
3  SELECT id, name, surname
4  FROM users
5  WHERE EXISTS (
6    SELECT user_id
7    FROM transaction1
8    WHERE transaction1.user_id = users.id
9    GROUP BY user_id
10   HAVING COUNT(id) > 80
11 );
12

```

The results pane shows 18 rows of data:

id	name	surname
185	Molly	Gillam
289	Dwight	Hivaru
318	Bryn	Achuw
454	Sfznoh	Xgvfrdxs
...	...	...

The bottom pane shows the query execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
110184	01:44:49	ALTER TABLE transaction1 ADD CONSTRAINT fr_transactions_companies FOREIGN K...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	4.813 sec
110185	01:49:40	USE users_transactions	0 row(s) affected	0.000 sec
110186	01:49:40	SELECT id, name, surname FROM users WHERE EXISTS (SELECT user_id FROM ...	4 row(s) returned	0.109 sec / 0.000 sec

- La consulta principal selecciona los usuarios que han realizado compras que cumplan los resultados de la subconsulta
- La subconsulta cuenta las transacciones de los usuarios que superan las 80 transacciones

Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

MySQL Workbench

Local instance MySQL80 x MySQL Model* x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

credit_card
transaction
Views
Stored Procedures
Functions
users_transactions
Tables
companies
credit_cards
transaction1
users
Views

Administration Schemas

Information

Schema: transactions

Query Editor

```

1 # Exercici 10
2
3 SELECT co.company_id, co.company_name, cc.iban, ROUND(AVG(tr.amount),2) AS mitjana_amount
4 FROM companies AS co
5 JOIN transaction1 AS tr ON co.company_id = tr.bussines_id
6 JOIN credit_cards AS cc ON tr.card_id = cc.id
7 WHERE co.company_name LIKE 'Donec Ltd'
8 GROUP BY co.company_id, co.company_name, cc.iban
9 ORDER BY mitjana_amount DESC;
10

```

Result Grid

company_id	company_name	iban	mitjana_amount
b-2242	Donec Ltd	US212768278745981809500233	708.14
b-2242	Donec Ltd	CA817443448587215122047864	690.81
b-2242	Donec Ltd	US157275989571099139868887	647.08
b-2242	Donec Ltd	US594988240619959562749388	638.82
b-2242	Donec Ltd	US186466262265569948042760	629.76

Result 19 x

Output

#	Time	Action	Message	Duration / Fetch
110185	01:49:40	USE users_transactions	0 row(s) affected	0.000 sec
110186	01:49:40	SELECT id, name, surname FROM users WHERE EXISTS (SELECT user_id FROM ...	4 row(s) returned	0.109 sec / 0.000 sec
110187	01:53:37	SELECT co.company_id, co.company_name, cc.iban, ROUND(AVG(tr.amount),2) AS mitj...	371 row(s) returned	0.125 sec / 0.000 sec

Query Completed

- Por legibilidad y coherencia (realidad económica), decido redondear a 2 decimales (con ROUND)
- Por estética y coherencia, ordeno los resultados de mayor a menor con ORDER BY

NIVELL 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

MySQL Workbench

Local instance MySQL80 x MySQL Model* x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

credit_card
transaction
Views
Stored Procedures
Functions
users_transactions
Tables
companies
credit_cards
transaction1
users
Views

Administration Schemas

Information

Schema: transactions

Query Editor

```

1 #NIVELL 2
2 #Exercici 1
3
4 SELECT DATE(tr.timestamp) AS data_venda, ROUND(SUM(tr.amount),2) AS total_vendes
5 FROM transaction1 AS tr
6 GROUP BY DATE(tr.timestamp)
7 ORDER BY total_vendes DESC
8 LIMIT 5;
9
10

```

Result Grid

data_venda	total_vendes
2019-11-18	15840.24
2023-02-20	15728.4
2022-12-13	15129.2
2019-03-18	14896.47
2017-12-20	14367.96

Result 21 x

Output

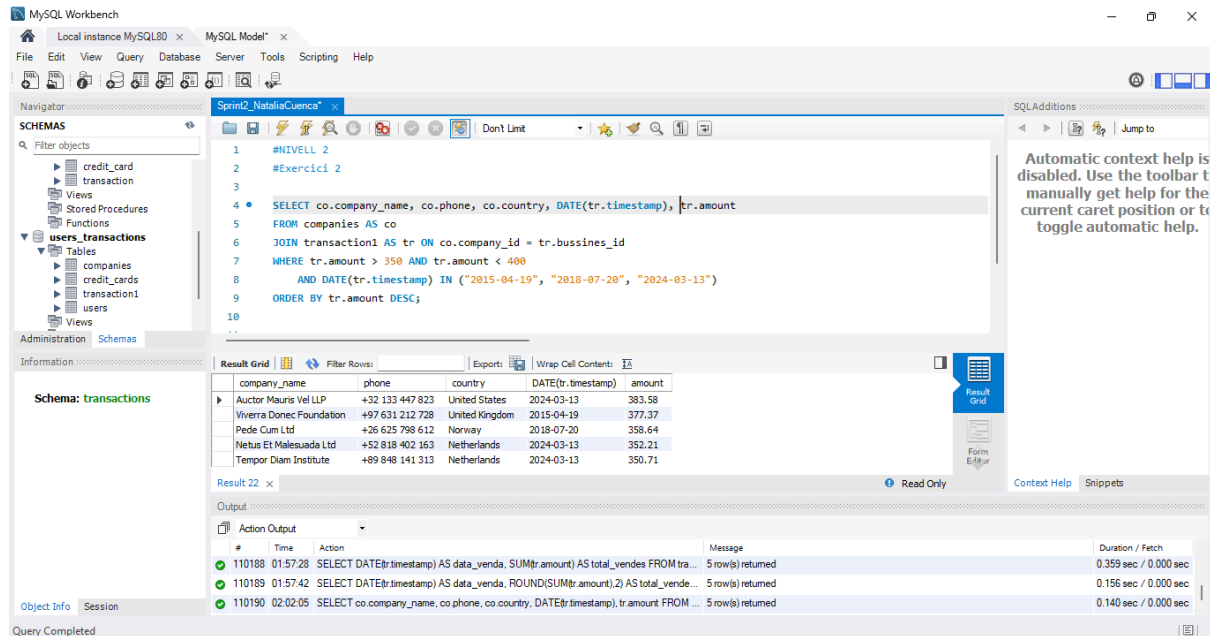
#	Time	Action	Message	Duration / Fetch
110187	01:53:37	SELECT co.company_id, co.company_name, cc.iban, ROUND(AVG(tr.amount),2) AS mitj...	371 row(s) returned	0.125 sec / 0.000 sec
110188	01:57:28	SELECT DATE(tr.timestamp) AS data_venda, SUM(tr.amount) AS total_vendes FROM tra...	5 row(s) returned	0.359 sec / 0.000 sec
110189	01:57:42	SELECT DATE(tr.timestamp) AS data_venda, ROUND(SUM(tr.amount),2) AS total_vende...	5 row(s) returned	0.156 sec / 0.000 sec

Query Completed

- Por legibilidad y coherencia económica, redondeo decimales con ROUND

Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.



MySQL Workbench interface showing the query execution for Exercise 2. The query is as follows:

```
1 #NIVELL 2
2 #Exercici 2
3
4 SELECT co.company_name, co.phone, co.country, DATE(tr.timestamp), tr.amount
5 FROM companies AS co
6 JOIN transaction1 AS tr ON co.company_id = tr.bussines_id
7 WHERE tr.amount > 350 AND tr.amount < 400
8 AND DATE(tr.timestamp) IN ("2015-04-19", "2018-07-20", "2024-03-13")
9 ORDER BY tr.amount DESC;
```

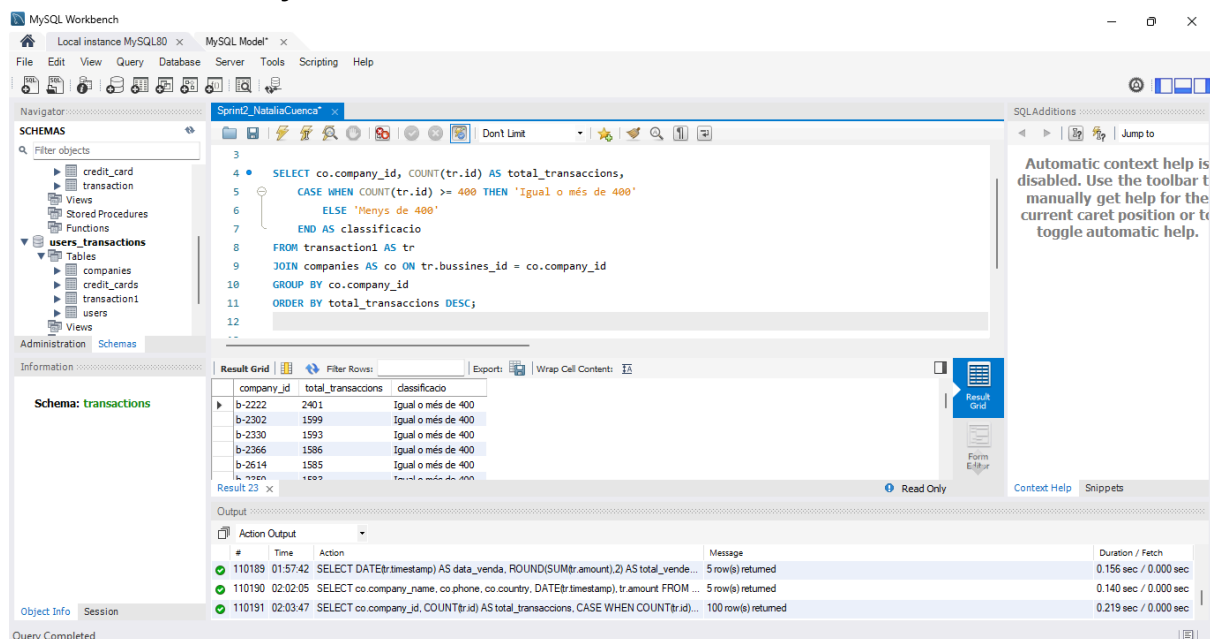
The result grid displays the following data:

company_name	phone	country	DATE(tr.timestamp)	amount
Auctor Mauris Vel LLP	+32 133 447 823	United States	2024-03-13	383.58
Viverra Donec Foundation	+97 631 212 728	United Kingdom	2015-04-19	377.37
Pede Cum Ltd	+26 625 798 612	Norway	2018-07-20	358.64
Netus Et Malesuada Ltd	+52 818 402 163	Netherlands	2024-03-13	352.21
Tempor Diam Institute	+89 848 141 313	Netherlands	2024-03-13	350.71

- En el enunciado entiendo que los valores 350 y 400 no deben estar incluidos, por eso empleo > y < en vez de BETWEEN

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.



MySQL Workbench interface showing the query execution for Exercise 3. The query is as follows:

```
3
4 SELECT co.company_id, COUNT(tr.id) AS total_transaccions,
5 CASE WHEN COUNT(tr.id) >= 400 THEN 'Igual o més de 400'
6 ELSE 'Menys de 400'
7 END AS classificacio
8 FROM transaction1 AS tr
9 JOIN companies AS co ON tr.bussines_id = co.company_id
10 GROUP BY co.company_id
11 ORDER BY total_transaccions DESC;
```

The result grid displays the following data:

company_id	total_transaccions	classificacio
b-2222	2401	Igual o més de 400
b-2302	1599	Igual o més de 400
b-2330	1593	Igual o més de 400
b-2366	1586	Igual o més de 400
b-2614	1585	Igual o més de 400
b-2621	1403	Igual o més de 400

Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

- Empiezo comprobando cuantos registros hay actualmente, que son 100000, tal como hemos visto anteriormente.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 #NIVELL 2
2 #Exercici 4
3
4 SELECT COUNT(id)
5 FROM transaction1;
6 # Salen 100000 registros
7
```

The Results tab shows the output of the query:

COUNT(id)
100000

The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
110190	02:02:05	SELECT co.company_name, co.phone, co.country, DATE(timestamp), tr.amount FROM ...	5 row(s) returned	0.140 sec / 0.000 sec
110191	02:03:47	SELECT co.company_id, COUNT(tr.id) AS total_transaccions, CASE WHEN COUNT(tr.id)...	100 row(s) returned	0.219 sec / 0.000 sec
110192	02:07:30	SELECT COUNT(id) FROM transaction1	1 row(s) returned	0.031 sec / 0.000 sec

- Localizo el que me interesa mediante LIKE

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

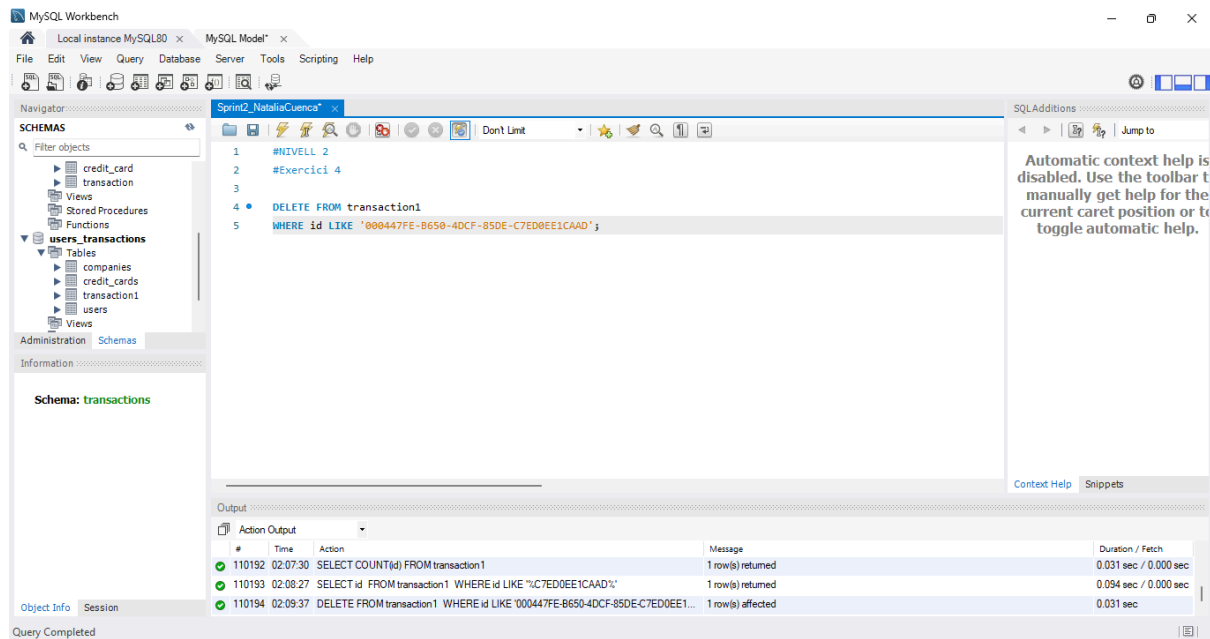
```
1 #NIVELL 2
2 #Exercici 4
3
4 SELECT id
5 FROM transaction1
6 WHERE id LIKE '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD%';
```

The Results tab shows the output of the query:

id
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD

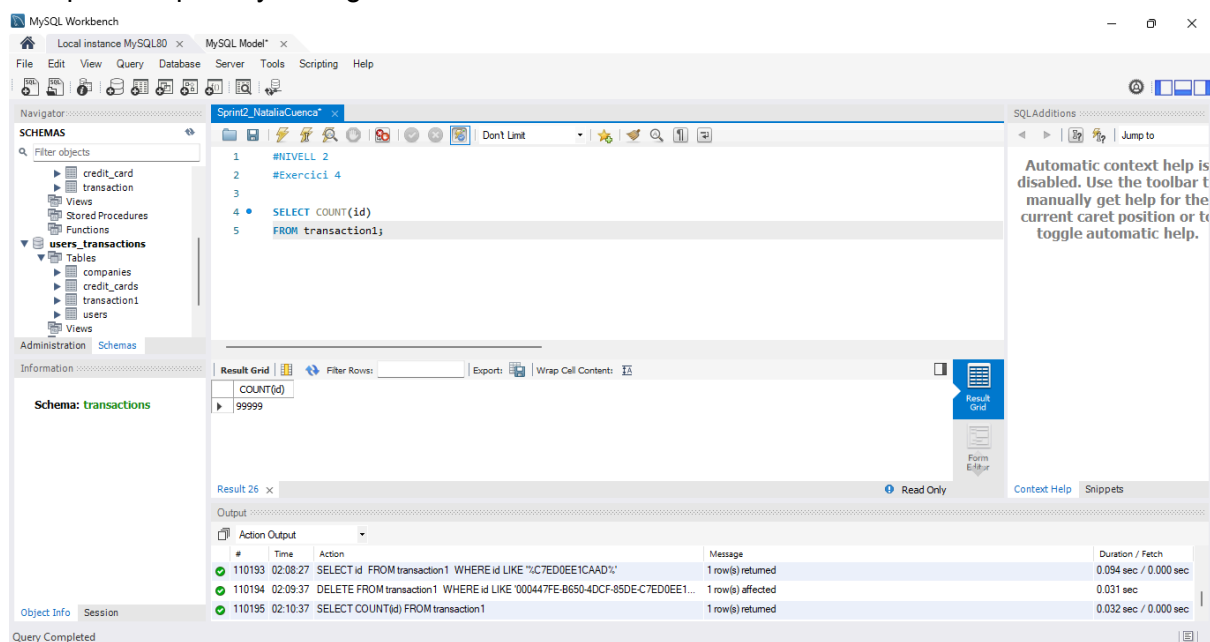
The Output tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
110191	02:03:47	SELECT co.company_id, COUNT(tr.id) AS total_transaccions, CASE WHEN COUNT(tr.id)...	100 row(s) returned	0.219 sec / 0.000 sec
110192	02:07:30	SELECT COUNT(id) FROM transaction1	1 row(s) returned	0.031 sec / 0.000 sec
110193	02:08:27	SELECT id FROM transaction1 WHERE id LIKE '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD%';	1 row(s) returned	0.094 sec / 0.000 sec

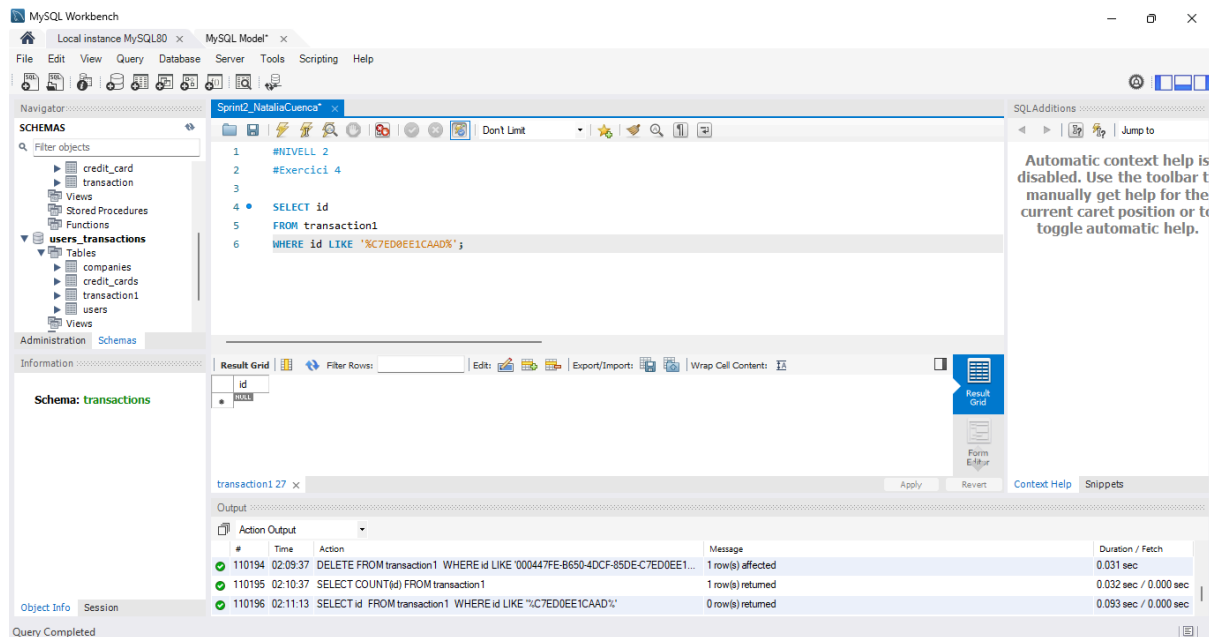


(en este caso no ha sido necesario desactivar temporalmente el filtro de seguridad con SET SQL_SAFE_UPDATES = 0; SET SQL_SAFE_UPDATES = 1;)

Compruebo que hay un registro menos:

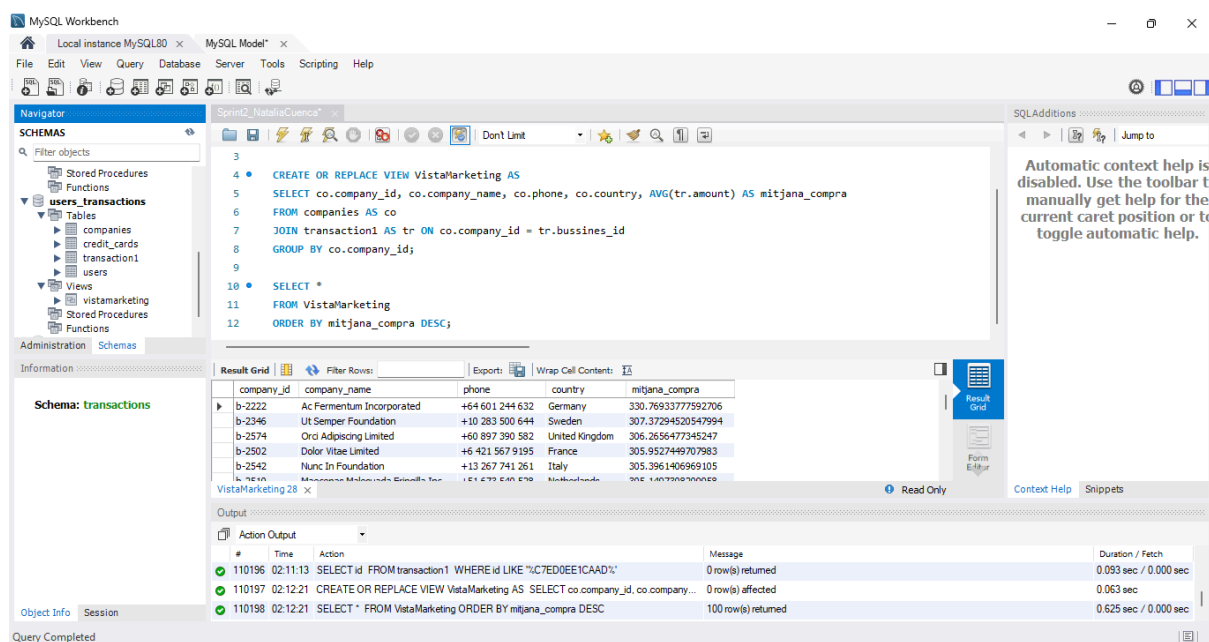


Otra comprobación puede ser mediante:



Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.



El diagrama actualizado con la vista:

